

Electronic Devices and Circuits (EE1004)

Date: September 23rd 2024

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 4

Roll No

Section

Student Signature

Do not write below this line

- Attempt all the questions.
- Show all the calculations, direct answers are not acceptable. Your answers should be approximated up to three decimal places.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

CLO #1: Analyze diode circuits and compute required parameters

[10 marks]

Q1:

Consider a half wave rectifier with capacitor filter fed by a 60Hz sinusoid having a peak value $V_p =$

100V. Let the load resistor $R = 10k\Omega$.

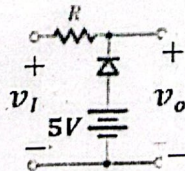
- Draw the circuit diagram and point out the output.
- Find the value of capacitor C that will result in a peak to peak ripple voltage of 1V.
- Illustrate the behavior of the circuit by drawing the output voltage waveform across capacitor including load.

CLO #1: Analyze diode circuits and compute required parameters

[10 marks]

Q2:

Illustrate the behavior of the given circuit by drawing the transfer characteristic graph. Use constant voltage drop model with $v_{D0} = 0.7V$ and $v_I = 15 \sin t$

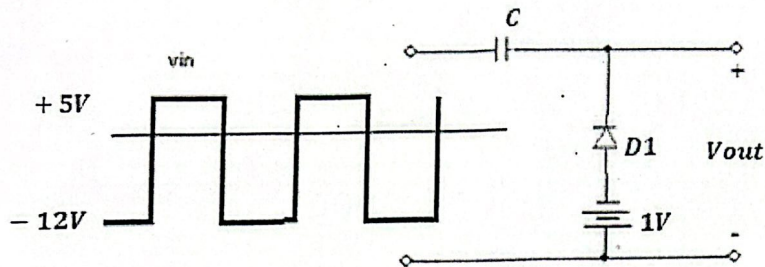


CLO #1: Analyze diode circuits and compute required parameters

Q3:

[10marks]

Analyze the following circuit



- Identify the name of this circuit?
- Find the maximum value of capacitor voltage?
- Illustrate the behavior of the circuit by sketching the output voltage waveform. Find the maximum and minimum value of the output.
- If we connect a resistor (in parallel with diode and 5V dc source) at the output, Sketch the voltage waveform.

CLO #1: Analyze diode circuits and compute required parameters

[10 marks]

Q4:

Sketch the output waveform to analyze the behavior of the given circuit. Use constant voltage drop model with $v_{D0} = 0.7V$ and $v_I = 20 \sin t$. Explain about the biasing of all diodes in positive and negative half cycle of the input.

